Lecture 22: Scaling up to GPT-3 and beyond

PSYC 51.17: Models of language and communication

Jeremy R. Manning
Dartmouth College
Winter 2026

Learning objectives

By the end of this lecture, you will be able to...

1. Describe how **GPT-2** demonstrated zero-shot task transfer through scale
2. Explain how **GPT-3** enabled **few-shot learning** without fine-tuning
3. Outline the alignment pipeline from RLHF to DPO to GRPO
4. Explain how **reasoning models** (o1, DeepSeek-R1) work and why they matter
5. Critically evaluate the debate over **emergent abilities** in LLMs

GPT-2: unsupervised multitask learning

The 10× scale-up

GPT-2 (Radford et al., 2019) scaled GPT-1 by an order of magnitude and discovered that **larger models can perform tasks without any fine-tuning**.

	GPT-1 (2018)	GPT-2 (2019)
Parameters	117M	1.5B (13×)
Training data	BooksCorpus (5B tokens)	WebText (40GB, ~8×)
Fine-tuning needed?	Yes	**No** (zero-shot transfer)

The key claim

"Language models are unsupervised multitask learners" — a single model trained on next-token prediction implicitly learns to translate, summarize, answer questions, and more. OpenAI staged the release over 9 months due to disinformation concerns — one of the first high-profile responsible AI releases.

Zero-shot task transfer

Performing tasks without fine-tuning

GPT-2 can perform tasks it was never explicitly trained for, simply by prompting with the right text format. This works because the model encountered similar patterns in its diverse training data.

Zero-shot prompting examples

Translation: `"English: I love machine learning\nFrench:"` → `"J'aime l'apprentissage automatique"`

Question answering: `"Q: What is the capital of France?\nA:"` → `"Paris"`

Summarization: `"[Long article]\n\nTL;DR:"` → generates a summary

The model learned these formats from web text where such patterns naturally occur (bilingual pages, Q&A forums, Reddit TL;DR summaries). Performance was far behind fine-tuned SOTA, but the fact that it worked *at all* was groundbreaking.

GPT-3: few-shot learning at scale

Brown et al. (2020): 'Language Models are Few-Shot Learners'

GPT-3 scaled up another **100×** (175B parameters, 300B tokens, ~$4.6M) and discovered **in-context learning** — the ability to learn new tasks from just a few examples in the prompt, with no gradient updates.

Few-shot sentiment classification

```
Classify each review as Positive or Negative.

Review: "This movie was amazing, I loved every minute!"
Sentiment: Positive

Review: "Terrible film, complete waste of time."
Sentiment: Negative

Review: "The acting was wooden and the plot made no sense."
Sentiment:
```

Beyond Chinchilla: the overtraining era

From Lecture 16's scaling laws to modern practice

Model	Parameters	Training tokens	Tokens/parameter	Strategy
GPT-3	175B	300B	1.7	Undertrained
Chinchilla	70B	1.4T	20	Compute-optimal
LLaMA 3	70B	15T	214	Inference-optimal
Phi-4	14B	10T+	700+	Extreme overtraining

The field has moved past Chinchilla-optimal. Since you train once but deploy

From GPT-3 to ChatGPT

The three-stage evolution

Stage	Model	Key innovation
1	GPT-3 (2020)	Next-token prediction at massive scale
2	InstructGPT (2022)	**Instruction tuning** on human-written responses
3	ChatGPT (Nov 2022)	**RLHF** (Lecture 16) for helpful, harmless conversation

Why ChatGPT mattered

ChatGPT reached **100M users in 2 months** — the fastest-growing consumer app in history. The key innovation wasn't the model (GPT-3.5) but the **alignment**: RLHF transformed a next-token predictor into a conversational assistant that *felt* helpful. This revealed that alignment technique matters as much as model scale.

Beyond RLHF: DPO and GRPO

Simpler, cheaper alternatives to RLHF

RLHF requires training a separate reward model and running PPO — complex and unstable. Two alternatives have emerged:

DPO (Rafailov et al., NeurIPS 2023): **Direct Preference Optimization** skips the reward model entirely. It directly optimizes the LLM to prefer winning responses over losing ones using a simple classification loss. Same quality as RLHF, ~3× less compute.

GRPO (Shao et al., 2024): **Group Relative Policy Optimization** from DeepSeekMath. Instead of human rankings, it generates multiple responses and uses a *verifiable reward* (e.g., "is the math answer correct?") to rank them. No humans needed for domains with checkable answers.

The trend

Alignment is getting **cheaper, simpler, and more automated**. This democratizes the ability to create instruction-following models — but also lowers the barrier for misuse.

Chain-of-thought prompting

Wei et al. (2022); Kojima et al. (2022)

Chain-of-thought (CoT) prompting asks the model to show its reasoning step by step before giving a final answer. This dramatically improves performance on reasoning tasks.

Chain-of-thought example

```
Q: Roger has 5 tennis balls. He buys 2 more cans of tennis
   balls. Each can has 3 balls. How many does he have now?

Let's think step by step:
1. Roger starts with 5 tennis balls
2. He buys 2 cans × 3 balls per can = 6 new balls
3. Total = 5 + 6 = 11 tennis balls

A: 11
```

Simply adding **"Let's think step by step"** improves accuracy on GSM8K math

Reasoning models: thinking before answering

A new paradigm: test-time compute scaling

Instead of making models bigger (training-time scaling), **reasoning models** spend more compute at inference time by generating an internal chain-of-thought before answering.

Key reasoning models

Model	Organization	Key innovation
o1 (2024)	OpenAI	Hidden chain-of-thought, trained with RL to "think"
o3 (2025)	OpenAI	Extended reasoning, SOTA on ARC-AGI
DeepSeek-R1 (2025)	DeepSeek	Open-weight, trained with GRPO, reasoning emerges from RL alone

DeepSeek-R1's surprising finding

R1 was trained with *pure RL* — no supervised reasoning examples. Yet it spontaneously developed chain-of-thought, self-correction, and even metacognition ("wait, let me reconsider..."). Then R1's reasoning was **distilled** into smaller models (Llama-70B, Qwen-32B), giving them reasoning abilities at a fraction of the cost.

Are emergent abilities real?

The debate

Wei et al. (2022) claimed that abilities like arithmetic and reasoning *emerge* suddenly at certain scales — absent in small models, present in large ones.

Schaeffer et al. (2023, NeurIPS) challenged this: they showed that "emergence" disappears when you switch from **nonlinear metrics** (exact-match accuracy) to **linear metrics** (token-level accuracy). The abilities were developing gradually all along — the metric just couldn't detect partial progress.

Why this matters for science

If emergence is real, it means we can't predict what larger models will do — they might develop unexpected and potentially dangerous capabilities. If it's a measurement artifact, then scaling is *predictable* and we can plan for it. The answer has profound implications for AI safety policy.

The modern LLM landscape (2025)

The field moves fast

Year	Milestone
2022	ChatGPT launched (OpenAI) — 100M users in 2 months
2023	GPT-4 (multimodal), Llama 2 (open weights), Claude 2
2024	Claude 3.5, GPT-4o, Llama 3, Mistral Large, Gemini 1.5
2024–25	Reasoning models (o1, o3, R1), small efficient models (Phi-4, Gemma 2)

Five key trends

1. **Reasoning at inference time** — spending more compute when thinking, not just when training
2. **Longer context windows** — 128K–1M+ tokens (Gemini 1.5 Pro)
3. **Open weight models** — LLaMA, Mistral, DeepSeek approaching frontier quality
4. **Multimodal** — text + vision + audio in a single model
5. **Efficiency** — smaller models matching larger predecessors (Phi-4 14B ≈ GPT-3.5 175B)

Current limitations of LLMs

What LLMs still struggle with

- **Hallucinations**: Confidently generate false statements with no awareness of uncertainty
- **Reasoning depth**: Multi-step logic and mathematical proofs remain unreliable (even reasoning models have limits)
- **Knowledge currency**: Training data has a cutoff date; no access to real-time information
- **Consistency**: May give different answers to the same question across runs
- **Prompt sensitivity**: Small wording changes can dramatically alter outputs

Active areas of research

Each limitation has spawned research directions: RAG for knowledge currency (Lecture 17), tool use and agents for grounding (Lecture 24), constitutional AI for alignment, reasoning models for reliability, and formal verification for correctness guarantees.

Discussion

Questions to consider

1. **Emergent abilities:** If Schaeffer is right that "emergence" is a measurement artifact, does that make scaling *more* or *less* concerning? What if Wei is right?
2. **Reasoning models:** DeepSeek-R1 developed chain-of-thought reasoning through pure RL — no human examples. Does this constitute "learning to think"? How is this different from how children learn to reason?
3. **The alignment tax:** DPO and GRPO make alignment cheaper and more accessible. Is this good (more aligned models) or dangerous (easier to create models aligned to harmful objectives)?
4. **Open vs closed:** DeepSeek-R1 is fully open. OpenAI's o1 is closed. If open models reach frontier quality, can safety-through-secrecy still

Further reading

Further reading

Brown et al. (2020, *NeurIPS*) "Language Models are Few-Shot Learners" — GPT-3: in-context learning at scale.

Rafailov et al. (2023, *NeurIPS*) "Direct Preference Optimization" — RLHF without the RL.

DeepSeek-AI (2025, *arXiv*) "DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning" — Open-weight reasoning model.

Schaeffer et al. (2023, *NeurIPS*) "Are Emergent Abilities of Large Language Models a Mirage?" — The metric artifact hypothesis.

Kojima et al. (2022, *NeurIPS*) "Large Language Models are Zero-Shot Reasoners" — "Let's think step by step."

OpenAI (2024, *arXiv*) "o1 System Card" — Reasoning via test-time compute scaling.

Questions?

[Email] [Discord] [Office hours]

Up next...

Implementing GPT from scratch: building a language model in PyTorch